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Convergence and Limitations of Fixed-Temperature Softmax Boltzmann Q-Learning

Table of Contents

Background and definitions

Visitation and exploration requirements for convergence

Fixed-temperature convergence guarantees and asymptotic error

Regret bounds and performance guarantees

Reward-scale and temperature sensitivity

Known pathologies and limitations

Background and definitions

Fixed-temperature softmax Boltzmann exploration picks actions with probabilities proportional to the exponential of their current Q-values divided by a temperature. The temperature sets the exploration–exploitation tradeoff: high temperature makes action choice closer to uniform, while low temperature makes it close to greedy. (6 sources)

In Q-learning, **Boltzmann exploration** usually means choosing action a in state s with probability proportional to \exp(Q_t(s,a)/T), where T>0 is a **temperature** parameter. This is the standard softmax action rule used to turn estimated action values into a stochastic policy. Early RL work already used this form, and its main role is to randomize action choice while still favoring actions with larger current Q-values. As T decreases, the policy puts more mass on the current best-valued action; in the limit T \to 0, it becomes greedy. As T becomes very large, the policy approaches near-uniform random choice. (Amin et al., 2021)

It helps to separate two related but different uses of “softmax” in the literature. One use is **action selection on top of ordinary Q-learning**, where the Q-update is the usual max-based backup, but behavior is sampled from a Boltzmann distribution over current Q-values. The other use is **soft Q-learning / entropy-regularized RL**, where the hard max in the Bellman operator is itself replaced by a soft-max or derived from an entropy bonus in the objective. Those methods are closely connected to maximum-entropy RL and use entropy regularization to delay collapse to deterministic policies and encourage exploration. (O'Donoghue, 2018) (Haarnoja et al., 2017) (Nachum et al., 2017) (Azar et al., 2010)

This distinction matters for convergence claims. For the user’s question, “fixed-temperature softmax Boltzmann exploration in Q-learning” typically refers to the **behavior policy** being Boltzmann with a constant temperature, not to solving an entropy-regularized control problem. In the entropy-regularized setting, the softmax policy can be the target optimum for a modified objective, while in ordinary Q-learning it is mainly an exploration mechanism layered onto value learning. (O'Donoghue, 2018) (Nachum et al., 2017) (Haarnoja et al., 2017)

A final piece of context is that Boltzmann exploration is widely used but theoretically subtle. Even though it is a classic and notable strategy, later theory work emphasizes that its behavior depends strongly on how the temperature or learning-rate schedule is chosen, and that understanding when it explores well is not automatic. This motivates the later discussion of visitation conditions, regret, reward scaling, and failure modes. (O'Donoghue, 2018) (Cesa-Bianchi et al., 2017)

Evidence

(Amin et al., 2021)

A Survey of Exploration Methods in Reinforcement Learning

"The most commonly used formalism for performing Softmax action selection is the Boltzmann distribution. The early use of Boltzmann distribution in exploration was by Watkins (1989); (Lin, 1992) and Barto et al. (1991). In the Boltzmann exploration approach, the value function Q t (s, a) for the current state S t = s and action a t = a assigns the probability of selecting action a as,

where T m > 0 is called the temperature and controls how frequent the agent will choose random actions as opposed to the best actions a t = arg max a Q t (s, a). If T m decreases, the probability of generating the action with the highest expected reward a t increases, leading to a decrease in the probability of selecting other actions and hence a lower probability of exploring the environment. In the limit of zero temperature T m → 0, the agent uses the target policy to select greedy actions in order to maximize the obtained rewards. At very large temperatures T m → "

(O'Donoghue, 2018)

Variational Bayesian Reinforcement Learning with Regret Bounds

"The update rule we derive is similar to that used in 'soft' Q-learning (so-called since the 'hard' max is replaced with a soft-max) (Azar et al., 2010)16,(Haarnoja et al., 2017)(Nachum et al., 2017)[47]. These approaches are very closely related to maximum entropy reinforcement learning techniques which add an entropy regularization 'bonus' to prevent early convergence to deterministic policies and thereby heuristically encourage exploration (Williams et al., 1991)59,(Mnih et al., 2016)(Sanghi, 2021)(Abdolmaleki et al., 2018)[25]. In our work the soft-max operator and entropy regularization arise naturally from the view of the agent as maximizing a risk-seeking exponential utility. Furthermore, in contrast to these other approaches, the entropy regularization is not a fixed hyper-parameter but something we explicitly control (or optimize for) in order to carefully trade-off exploration and exploitation"

". Following a Boltzmann policy over the K-values yields a practical algorithm that we call 'K-learning' which attains a Bayes regret upper bounded byÕ(L ✓ SAT) 1 where L is the time horizon, S is the total number of states, A is the number of actions per state, and T is the number of elapsed timesteps (Cesa-Bianchi et al., 2017). This regret bound matches the best known bound for Thompson sampling up to log factors (Osband et al., 2013) and is within a factor of ✓ L of the known information theoretic lower bound"

(Haarnoja et al., 2017)

Reinforcement Learning with Deep Energy-Based Policies

"We propose a method for learning expressive energy-based policies for continuous states and actions, which has been feasible only in tabular domains before. We apply our method to learning maximum entropy policies, resulting into a new algorithm, called soft Q-learning, that expresses the optimal policy via a Boltzmann distribution. We use the recently proposed amortized Stein variational gradient descent to learn a stochastic sampling network that approximates samples from this distribution. The benefits of the proposed algorithm include improved exploration and compositionality that allows transferring skills between tasks, which we confirm in simulated experiments with swimming and walking robots. We also draw a connection to actor-critic methods, which can be viewed performing approximate inference on the corresponding energy-based model."

(Nachum et al., 2017)

Bridging the Gap Between Value and Policy Based Reinforcement Learning

"We establish a new connection between value and policy based reinforcement learning (RL) based on a relationship between softmax temporal value consistency and policy optimality under entropy regularization. Specifically, we show that softmax consistent action values correspond to optimal entropy regularized policy probabilities along any action sequence, regardless of provenance. From this observation, we develop a new RL algorithm, Path Consistency Learning (PCL), that minimizes a notion of soft consistency error along multi-step action sequences extracted from both on- and off-policy traces. We examine the behavior of PCL in different scenarios and show that PCL can be interpreted as generalizing both actor-critic and Q-learning algorithms. We subsequently deepen the relationship by showing how a single model can be used to represent both a policy and the corresponding softmax state values, eliminating the need for a separate critic. The experimental evaluation demonstrates that PCL significantly outperforms strong actor-critic and Q-learning baselines across several benchmarks."

(Azar et al., 2010)

Dynamic policy programming

"In this paper, we propose a novel policy iteration method, called dynamic policy programming (DPP), to estimate the optimal policy in the infinite-horizon Markov decision processes. We prove the finite-iteration and asymptotic l\infty-norm performance-loss bounds for DPP in the presence of approximation/estimation error. The bounds are expressed in terms of the l\infty-norm of the average accumulated error as opposed to the l\infty-norm of the error in the case of the standard approximate value iteration (AVI) and the approximate policy iteration (API). This suggests that DPP can achieve a better performance than AVI and API since it averages out the simulation noise caused by Monte-Carlo sampling throughout the learning process. We examine this theoretical results

numerically by com- paring the performance of the approximate variants of DPP with existing reinforcement learning (RL) methods on different problem domains. Our results show that, in all cases, DPP-based algorithms outperform other RL methods by a wide margin."

(Cesa-Bianchi et al., 2017)

Boltzmann Exploration Done Right

"Despite its widespread use, there is virtually no theoretical understanding about the limitations or the actual benefits of this exploration scheme. Does it drive exploration in a meaningful way? Is it prone to misidentifying the optimal actions or spending too much time exploring the suboptimal ones? What is the right tuning for the learning rate?"

"One of our main results is showing that the Boltzmann exploration strategy with any monotone learning-rate sequence will induce suboptimal behavior. As a remedy, we offer a simple non-monotone schedule that guarantees near-optimal performance, albeit only when given prior access to key problem parameters that are typically not available in practical situations (like the time horizon T and the suboptimality gap Δ). More importantly, we propose a novel variant that uses different learning rates for different arms, and achieves a distribution-dependent regret bound of order $\frac{K\log^2 T}{\Delta}$ and a distribution-independent bound of order $\sqrt{KT}\log K$ without requiring such prior knowledge"

".We first show that any monotone learning-rate schedule will inevitably force the policy to either spend too much time drawing suboptimal arms or completely fail to identify the optimal arm. Then, we show that a specific non-monotone schedule of the learning rates can lead to regret bound of order K log T Δ 2 . However, the learning schedule has to rely on full knowledge of the gap Δ and the number of rounds T . Moreover, our negative result helps us to identify a crucial shortcoming of the Boltzmann exploration policy: it does not reason about the uncertainty of the empirical reward estimates."

Visitation and exploration requirements for convergence

For standard tabular Q-learning, the core requirement is still that every state–action pair must be updated infinitely often. A fixed-temperature softmax policy can help keep all actions sampled, but convergence is not guaranteed just from using softmax; the real issue is whether the induced sampling process actually delivers enough visitation. (7 sources)

The basic convergence condition for tabular Q-learning is the classic one: all state–action pairs must keep being updated, in the limit infinitely often. Later summaries restate this plainly: convergence to the optimal policy or optimal Q^* is proved only under continued updates of all state–action pairs, and this is exactly the condition that is often hard to satisfy in practice. (Grana et al., 2011) (Choshen et al., 2018)

This is why the literature often moves from “use softmax” to stronger exploration conditions such as GLIE, where exploration persists long enough but vanishes in the limit so the policy becomes greedy. Grana et al. note that GLIE policies were proposed precisely to relax the practical difficulty of guaranteeing sufficient visitation under naive exploration. Related applied work with Boltzmann exploration therefore typically uses a decaying temperature, not a fixed one, so that action choice starts exploratory and becomes more exploitative over time; Ceran et al. explicitly describe the temperature as decaying over iterations and cite convergence results under such schedules. (Grana et al., 2011) (Ceran et al., 2021) (Ceran et al., 2019)

For Boltzmann exploration specifically, the important point is that softmax probabilities are only useful for convergence if they imply enough actual visits. In finite tabular problems, convergence results for dynamic Boltzmann policies are still stated with the same visitation condition: all state–action pairs must be visited infinitely often. In other words, Boltzmann action selection does not remove the need for exhaustive asymptotic sampling; it is just one way to try to achieve it. (Pan et al., 2019)

A fixed positive temperature can, in principle, keep every action at every visited state at nonzero probability, which helps avoid complete collapse to a single action. But that alone is not a full convergence guarantee, because state visitation also depends on the environment dynamics, and because the probability of poor actions can become extremely small when Q-gaps grow. So the practical question is not only whether each action has nonzero softmax mass, but whether the resulting on-policy trajectory still revisits every state–action pair often enough. This broader difficulty is one reason modern theory treats time-varying on-policy learning as subtle. (Model-Generated) (Nanda et al., 2025)

Recent theory makes this point from a different angle: analyses for stationary behavior policies often rely on Markov-chain mixing assumptions, but for time-varying policies such as \varepsilon-greedy or Boltzmann

exploration it is not clear how to apply those tools without strong uniform ergodicity assumptions over every policy encountered. Nanda et al. highlight that such assumptions can be too strong and can hide the actual exploration–exploitation tradeoff; they instead assume only that there exists some policy that induces an irreducible state Markov chain. This shows that the main challenge is not “softmax versus not softmax” by itself, but how the sampling policy affects coverage and mixing over time. (Nanda et al., 2025) (Zou et al., 2019)

Finally, even when softmax gives more informed exploration than a random walk, it is not directed toward poorly known parts of the MDP. Choshen et al. stress that Boltzmann exploration favors actions with higher learned value, but still does not specifically bias toward unvisited trajectories. So for convergence, fixed-temperature softmax should be viewed as a heuristic that may maintain stochastic action choice, not as a substitute for the formal infinite-visitation requirement. (Choshen et al., 2018)

Evidence

(Grana et al., 2011)

Cooperative Multi-Agent Reinforcement Learning for Multi-Component Robotic Systems: guidelines for future research

> *"In (Watkins et al., 2004) convergence of Q-learning in a stationary environment to an optimal policy with probability 1 is proved, as long as all state-actions pairs keep being updated. This is a theoretically sensible condition, but hard to meet in practice because an agent may not be able to explore sufficient space to guarantee convergence. To relax this condition, Greedy in the Limit with Infinite Exploration (GLIE) policies were proposed [61]."*

(Choshen et al., 2018)

DORA The Explorer: Directed Outreaching Reinforcement Action-Selection

> *"Formally, to guarantee convergence to Q * , the Q-Learning algorithm must visit each state-action pair infinitely many times. A naive random walk exploration is sufficient for converging asymptotically. However, such random exploration has two major limitations when the learning process is finite. First, the agent would not utilize its current knowledge about the world to guide its exploration. For example, an action with a known disastrous outcome will be explored over and over again. Second, the agent would not be biased in favor of exploring unvisited trajectories more than the visited ones -hence "wasting" exploration resources on actions and trajectories which are already well known to it"*

> *".Another popular alternative, emphasizing the preference to learn about actions associated with higher rewards, is to draw actions from a Boltzmann Distribution (Softmax) over the learned Q values, regulated by a Temperature parameter. While such approaches lead to more informed exploration that is based on learning experience, they still fail to address the second issue, namely they are not directed (Thrun, 1992) towards gaining more knowledge, not biasing actions in the direction of unexplored trajectories."*

(Ceran et al., 2021)

Learning to Minimize Age of Information over an Unreliable Channel with Energy Harvesting

> *"Parameter τ n in (14) is called the temperature parameter and decays exponentially with decay parameter γ τ ≤ 1 at each iteration. High τ n corresponds to more uniform action selection (exploration) whereas low τ n is biased toward the best action (exploitation). According to Theorem 2 of (Gosavi, 2004), if α, β satisfy"*

> *"α(m) → 0, GR-Learning converges to an optimal policy."*

(Ceran et al., 2019)

Reinforcement Learning to Minimize Age of Information with an Energy Harvesting Sensor with HARQ and Sensing Cost

> *"The transmitter action selection method should balance the exploration of new actions with the exploitation of actions known to perform well. In particular, the Boltzmann (softmax) action selection method, which chooses each action randomly relative to its expected cost, is used in this paper as follows:"*

> *"Parameter τ in (12) is called the temperature parameter and decays exponentially with decay parameter γ. High τ corresponds to more uniform action selection (exploration) whereas low τ is biased toward the best action (exploitation). According to Theorem 2 of (Gosavi, 2004), if α, β satisfy"*

> *"α(m) → 0, GR-Learning converges to an optimal policy."*

(Pan et al., 2019)

Reinforcement Learning with Dynamic Boltzmann Softmax Updates

"Theorem 3 (Convergence of DBS Q-learning) The Q-learning algorithm with dynamic Boltzmann softmax policy given by"

*"converges to the optimal Q * (s, a) values if 1. The state and action spaces are finite, and all state-action pairs are visited infinitely often."*

(Nanda et al., 2025)

A Minimal-Assumption Analysis of Q-Learning with Time-Varying Policies

"In practice, however, Q-learning is almost always implemented with time-varying policies, such as -greedy (Mnih et al., 2015), Boltzmann (softmax) exploration (Leslie et al., 2005)41,(Meyn, 2024), or combinations and variants of these (Haarnoja et al., 2017) (Hasselt et al., 2015)[58]"

".Existing analyses of RL algorithms under stationary learning policies typically rely on Markov chain mixing arguments (Bhandari et al., 2018)(Srikant et al., 2019). However, when the policy is time-varying, it is unclear how to apply such techniques without imposing strong assumptions-such as requiring every policy encountered by the algorithm's trajectory to induce a uniformly ergodic Markov chain with mixing rates uniformly bounded from above and stationary distributions uniformly bounded away from zero [41](Zou et al., 2019). Moreover, under such assumptions, one cannot theoretically capture the exploration-exploitation trade-off inherent in Q-learning with on-policy sampling"

".Specifically, we assume only the existence of a policy that induces an irreducible Markov chain over states. This policy need not be encountered along the algorithm's trajectory and can therefore be viewed as a mild, algorithm-independent assumption on the underlying stochastic model. Under this assumption, we establish last-iterate convergence rates for on-policy Q-learning that quantitatively characterize the explorationexploitation trade-off."

(Zou et al., 2019)

Finite-Sample Analysis for SARSA with Linear Function Approximation

"SARSA is an on-policy algorithm to learn a Markov decision process policy in reinforcement learning. We investigate the SARSA algorithm with linear function approximation under the non-i.i.d.\ setting, where a single sample trajectory is available. With a Lipschitz continuous policy improvement operator that is smooth enough, SARSA has been shown to converge asymptotically. However, its non-asymptotic analysis is challenging and remains unsolved due to the non-i.i.d. samples, and the fact that the behavior policy changes dynamically with time. In this paper, we develop a novel technique to explicitly characterize the stochastic bias of a type of stochastic approximation procedures with time-varying Markov transition kernels. Our approach enables non-asymptotic convergence analyses of this type of stochastic approximation algorithms, which may be of independent interest. Using our bias characterization technique and a gradient descent type of analysis, we further provide the finite-sample analysis on the mean square error of the SARSA algorithm. In the end, we present a fitted SARSA algorithm, which includes the original SARSA algorithm and its variant as special cases. This fitted SARSA algorithm provides a framework for \textit{iterative} on-policy fitted policy iteration, which is more memory and computationally efficient. For this fitted SARSA algorithm, we also present its finite-sample analysis."

Fixed-temperature convergence guarantees and asymptotic error

*With a fixed nonzero softmax temperature, ordinary Q-learning generally does **not** have the same "converges to the exact greedy-optimal \(Q^*\)" guarantee that decaying-temperature GLIE schemes do. What you can often guarantee instead is convergence to a **soft / interior limit** or to a **bounded neighborhood** of \(Q^*\), with the size of that neighborhood shrinking as temperature is lowered in the right parameterization. (5 sources)*

A key distinction in the literature is between **decaying-temperature** Boltzmann exploration and **fixed-temperature** Boltzmann exploration. Convergence-to-optimality results are standard for GLIE-type policies, which include Boltzmann rules with a temperature that decreases over time, but these results do not directly transfer to a constant temperature. Smirnova et al. explicitly contrast the two: they note convergence to optimality for SARSA with GLIE policies that include decaying-temperature Boltzmann policies, while also emphasizing that with a **fixed** temperature the relevant guarantee is instead a robustness/performance tradeoff, not exact optimality of the standard RL objective (Smirnova et al., 2019).

For fixed-temperature soft operators, the asymptotic limit is usually a **softened solution** rather than the hard-max optimum. This is clearest in entropy-regularized or soft-Bellman settings: with a fixed smoothing parameter, the operator converges to its own fixed point, and only in the limit as the inverse temperature grows without bound does that solution approach the usual max-based optimum (Hu et al., 2021) (Kim et al., 2019). That same intuition carries over to action-selection schemes layered on Q-learning: if the temperature stays positive, the policy does not become greedy in the limit, so one should not expect exact convergence to

the standard deterministic-optimal control solution without extra arguments.

Recent analysis also makes the guarantee more precise for the **Boltzmann softmax operator** itself. Lee et al. show that, unlike max, log-sum-exp, or mellowmax, the Boltzmann softmax operator is generally **not non-expansive** and the induced Bellman map may fail to be a contraction in the \cdot*infty norm. Because of this, the usual contraction proof for exact convergence does not apply. Under i.i.d. sampling, diminishing step sizes, and a full-support visitation assumption d(s,a)>0 for all state-action pairs, they prove almost-sure convergence only to an **invariant error set** around Q^*{\max}, not necessarily to Q^*_{\max} itself (Lee et al., 2024). Their bound makes the temperature effect explicit: the radius scales like \gamma \ln(|A|)/((1-\gamma)\lambda), so the asymptotic error gets smaller as the inverse temperature \lambda increases, i.e., as the softmax becomes sharper (Lee et al., 2024).

This yields a useful practical reading of fixed-temperature guarantees: a larger \lambda (lower temperature) can reduce asymptotic bias relative to the hard-max solution, but then behavior becomes closer to greedy; a smaller \lambda gives more smoothing and more tolerance to update noise or approximation error, but also larger asymptotic deviation from Q^*. Smirnova et al. summarize exactly this point for fixed-temperature softmax-style value methods: they are tolerant to errors, and the temperature controls the tradeoff between asymptotic performance and error sensitivity (Smirnova et al., 2019).

There is also an older but notable line of results from learning dynamics in games showing that any finite nonzero exploration rate leads to convergence to an **interior rest point** rather than to a pure greedy corner solution, and that changing the exploration rate can even change the number and stability of rest points; above a critical exploration level there may be a unique globally stable rest point (Kianercy et al., 2011). While this is not a tabular Q-learning proof for MDP control, it reinforces the same high-level message: fixed exploration can stabilize the dynamics, but the stabilized limit is generally a **soft mixed solution**, not exact greedy optimality (Kianercy et al., 2011).

Evidence

(Smirnova et al., 2019)

On the Convergence of Approximate and Regularized Policy Iteration Schemes

> *"(Singh et al., 2000) proves convergence to optimality of the SARSA algorithm with GLIE policies ("greedy in the limit with infinite exploration") that include a class of Boltzmann policies with decaying temperature. Another close work [18] studies convergence to optimality of a value iteration algorithm with a dynamic Boltzmann operator that represents an instance of the reg-MPI scheme with m = 1, negative entropy regularizer and decreasing temperature schedule"*
>
> *". The authors showed that (1) value-based algorithms using softmax operator and fixed temperature parameter are tolerant to any type of errors, e.g. arising from function approximation or due to finite sample of observations used to perform the updates;*
>
> *(2) the temperature parameter controls the trade-off between the asymptotic performance and the sensitivity to errors."*

(Hu et al., 2021)

Count-Based Temperature Scheduling for Maximum Entropy Reinforcement Learning

> *"In early stages of training, β should intuitively be assigned a small value to induce stochasticity, and in later stages of training β → ∞ to approach the deterministic behavior that optimizes the standard reinforcement learning objective"*
>
> *".Mellowmax is non-decreasing in β (Kim et al., 2019) and a non-expansion for a fixed β under the supremum norm (Fox et al., 2016), and as β → ∞ it converges to pure maximization"*
>
> *".When the Q function has a tabular representation, the contraction property of the soft Bellman operator guarantees convergence to the operator's fixed point, the softmax of which is the optimal policy in (1) (Fox et al., 2016)."*

(Kim et al., 2019)

DeepMellow: Removing the Need for a Target Network in Deep Q-Learning

> *"Deep Q-Network (DQN) is an algorithm that achieves human-level performance in complex domains like Atari games. One of the important elements of DQN is its use of a target network, which is necessary to stabilize learning. We argue that using a target*

network is incompatible with online reinforcement learning, and it is possible to achieve faster and more stable learning without a target network when we use Mellowmax, an alternative softmax operator. We derive novel properties of Mellowmax, and empirically show that the combination of DQN and Mellowmax, but without a target network, outperforms DQN with a target network."

(Lee et al., 2024)

Toward a Unified Lyapunov-Certified ODE Convergence Analysis of Smooth Q-Learning with p-Norms

"Assumption 2. d(s, a) > 0 holds for all (s, a) ∈ S × A"

".Assumption 2 ensures sufficient exploration by guaranteeing that every state-action pair is visited infinitely often"

".In contrast, (Beck et al., 2012) imposes an alternative exploration requirement, known as the cover time condition, which posits that within a prescribed time horizon every state-action pair is expected to be visited at least once. Related variants of cover-time conditions have also been used in (Even-Dar et al., 2004) and [23] to derive convergence-rate guarantees"

*".3. A non-contractive stability result for Boltzmann smooth Q-learning. For the Boltzmann softmax operator, which is generally not non-expansive, we prove that the corresponding ODE converges to an explicit error set around Q * max . The radius of this set scales as γ ln(|A|)/((1 − γ)λ), showing how the approximation error shrinks as the temperature parameter λ increases"*

".4. A unified asymptotic convergence theorem for the stochastic algorithms. Using the ODE stability results, we derive almost-sure convergence guarantees for the corresponding discrete-time Q-learning variants under i.i.d. sampling and standard diminishing step-sizes"

".For the Boltzmann softmax operator, the induced Bellman operator need not be contractive. In this case, the ODE and the stochastic iterates are guaranteed to approach the invariant error set"

"Consequently, the limiting error shrinks as the temperature parameter λ increases; see Appendix Section C and Theorem 6"

*".Since the Boltzmann softmax operator is, in general, not non-expansive with respect to the ∞-norm (Asadi et al., 2016), the induced mapping F may fail to be a contraction in the ∞-norm. Consequently, the contraction-based approach used for the max, LSE, and mellowmax operators does not apply directly. Instead, we show that Algorithm 1 with the Boltzmann softmax operator converges to a bounded neighborhood of Q * max . Theorem 6. Let us assume that the step-sizes satisfy"*

"Moreover, let us consider the Boltzmann softmax operator. Then, the corresponding Algorithm 1 satisfies dist(Q k , H) → 0 with probability one"

(Kianercy et al., 2011)

Dynamics of Boltzmann Q learning in two-player two-action games.

"for any finite (non-zero) exploration rate, the learning dynamics necessarily converges to an interior rest point. This seems to be in contrast with previous observation (Tuyls et al., 2005), where we believe the authors have confused slow convergence with limit cycles"

".depending on the game, there can be one, two, or three rest points, with a bifurcation between different rest-point structures as one varies the exploration rate. In particular, there is a critical exploration rate above which there remains only one rest point, which is globally stable."

Regret bounds and performance guarantees

*Regret guarantees for Boltzmann exploration mostly exist for **carefully scheduled, usually decaying-temperature** variants, not for naive fixed-temperature softmax. When theory is available, performance depends strongly on the temperature schedule, the MDP gap, and how much exploration is kept; fixed-temperature softmax by itself does not come with a strong general regret guarantee. (3 sources)*

On the regret side, the cleanest positive results in this family usually do **not** apply to plain fixed-temperature Boltzmann Q-learning. Instead, they apply to algorithms that still use a Boltzmann policy but add extra structure, such as optimism and an explicit inverse-temperature schedule. A clear example is **K-learning**: O'Donoghue et al. show that following a Boltzmann policy over their K-values gives a **Bayesian regret bound** matching Thompson sampling up to logarithmic factors under their assumptions. But they also stress that this is **not** just ordinary soft Q-learning: the method uses a planned schedule for the inverse temperature and an optimistic value construction, and those changes are what make the exploration strategy principled. (O'Donoghue et al., 2020)

For Boltzmann **Q-learning** itself, available regret results again point to the importance of a **decaying** temperature rather than a fixed one. Singh et al. analyze Boltzmann Q-learning with decaying temperature and show that regret depends critically on the **suboptimality gap** of the MDP: if gaps are large enough, regret can be sublinear, but when gaps are small it worsens and can get close to linear. They also show that the temperature schedule affects both convergence speed and the induced mixing behavior of the on-policy sampling process, and note that the temperature decay parameter must scale as O(1-\gamma) to keep enough exploration. Their guarantees are uniform in time, but they are for this scheduled-exploration setting, not for an arbitrary constant-temperature rule. (Singh et al., 2026)

Older bandit results tell a similar story. Kuleshov et al. note that, as with \varepsilon-greedy, **polylogarithmic regret bounds** exist for **decreasing-temperature** softmax schedules, while their experiments instead use fixed temperatures for practical reasons. This is useful context: the theory literature has long had regret guarantees for annealed softmax schedules, but that does **not** imply a comparable bound for constant-temperature Boltzmann exploration. (Kuleshov et al., 2014)

So the practical takeaway is narrow but important. If the question is whether **fixed-temperature softmax Boltzmann exploration in ordinary Q-learning** has a general no-regret or strong finite-time optimality guarantee, the answer is basically **no known broad guarantee from these sources**. The positive guarantees are mainly for: (i) decreasing-temperature schedules, often with gap-dependent performance, or (ii) modified Boltzmann-style algorithms such as K-learning that add optimism and carefully tune the temperature over time. (O'Donoghue et al., 2020) (Singh et al., 2026) (Kuleshov et al., 2014)

Evidence

(O'Donoghue et al., 2020)

Making Sense of Reinforcement Learning and Probabilistic Inference

> *"Following a Boltzmann policy over these K-values satisfies a Bayesian regret bound which matches the current best bound for Thompson sampling up to logarithmic factors under the same set of assumptions"*
>
> *".They both solve a 'soft' value function and use Boltzmann policies. However, the differences are important. Firstly, K-learning has an explicit schedule for the inverse temperature parameter β = β ✓ , and secondly it replaces the expected reward (with respect to the posterior) with a quantity that is optimistic for the expected reward under the posterior. These two relatively small changes make K-learning a principled exploration and inference strategy"*
>
> *".As β ∞ K-learning converges on pulling arm 2 with probability one. This is in contrast to soft Q-learning where arm 2 is exponentially unlikely to be selected as the exploration parameter β grows."*

(Singh et al., 2026)

Regret and Sample Complexity of Online Q-Learning via Concentration of Stochastic Approximation with Time-Inhomogeneous Markov Chains

> *"We first analyze Boltzmann Q-learning with decaying temperature and show that its regret depends critically on the suboptimality gap of the MDP: for sufficiently large gaps, the regret is sublinear, while for small gaps it deteriorates and can approach linear growth"*
>
> *".Our results show how the exploration parameters, specifically the temperature schedule for the softmax exploration and the sequence n for greedy policies, affect the convergence rate of the learned Q-values to Q Moreover, our concentration bounds enable us to establish almost sure convergence of the Q-learning iterates, which is nontrivial for online Q-learning with decaying exploration"*
>
> *".Condition (a) controls the rate at which the contraction factor approaches one and can be interpreted as a constraint on the growth of the mixing time, since the previous assumptions imply μ min (ζ n) ≥ c 3 /(c 2 (n + n 0) κ1)"*
>
> *".While the divergence of the mixing time is intrinsic to vanishing exploration in online Q-learning, we emphasize that the growth of the Lipschitz constant L 2 (ζ) is also unavoidable for softmax-based exploration. As the temperature decays to promote exploitation, the induced transition kernel becomes increasingly sensitive to changes in the iterate"*
>
> *".The convergence rate of Boltzmann Q-learning improves as a and b decrease, and can be made arbitrarily close to Õ(1/ ✓ n) by choosing a = 0 (corresponding to a stepsize of order Θ(1/n)) and taking b sufficiently small. Smaller values of b correspond to larger*

temperatures and hence increased exploration"

".Finally, we note that b must scale as O(1 − γ) to ensure sufficient exploration"

".Intuitively, gap(M) is the smallest difference between the value of an optimal action and a suboptimal action, over all states. Smaller gaps correspond to harder problems, as optimal and suboptimal actions are harder to distinguish"

".We emphasize that our regret guarantees are uniform in time, meaning they hold for all N and do not require prior knowledge of the horizon."

(Kuleshov et al., 2014)

Algorithms for multi-armed bandit problems

"As in the case of -greedy, polylogarithmic regret bounds exist Cesa-Bianchi and Fischer (1998) for decreasing τ schedules. However, because empirical evidence suggests that such schedules offer no practical advantage (Vermorel et al., 2005), we will use fixed values of τ in our experiments."

Reward-scale and temperature sensitivity

Fixed-temperature softmax is very sensitive to the numerical scale of the Q-values and rewards, because the policy depends on exponentials of \(Q/T\). In practice, changing the reward scale without retuning temperature can make the policy act much more greedy or much more random, and good asymptotic behavior usually requires a decaying temperature schedule rather than a fixed one. (3 sources)

A central practical issue with Boltzmann exploration is that the temperature only has meaning **relative to the scale of the Q-values**, and hence relative to the reward scale. If rewards are multiplied up, the gaps between Q-values also tend to grow, so a fixed temperature makes the softmax much sharper; if rewards are shrunk, the same temperature makes action choice flatter. Put differently, fixed-temperature softmax is not reward-scale invariant: the same problem, written with a different reward magnitude, can induce very different exploration behavior unless temperature is retuned. This is one reason the method is often described as sensitive in practice. (Model-Generated)

The cited theory supports this gap-based view directly. Guo et al. stress that Boltzmann policies are smooth, unlike discontinuous argmax-based selection, but also that the performance gap between the Boltzmann policy and the optimal one depends on choosing the softmax hyper-parameter appropriately. Their analysis is driven by the **action gap** between the best and second-best actions, because one needs enough separation to distinguish optimal from suboptimal choices; when that gap becomes small, control gets harder. (Guo et al., 2019) (Bellemare et al., 2015)

This action-gap view explains the reward-scale sensitivity. Softmax probabilities depend on relative score differences, so if all Q-gaps are small compared with temperature, the policy stays diffuse; if Q-gaps are large compared with temperature, the policy becomes nearly greedy. Since reward rescaling changes those effective gaps, it changes the exploration level even when the nominal temperature is unchanged. Guo et al. also note that one must control the difference between optimal and non-optimal policies through the softmax parameter, which is another way of saying that temperature tuning is problem-scale dependent. (Guo et al., 2019)

There is also a stability tradeoff. A smoother Boltzmann rule can avoid the instability that comes from the discontinuity and sensitivity of argmax-style choice, but this same smoothing introduces bias away from the optimal greedy policy unless the temperature is set sharply enough. So temperature is doing two jobs at once: it smooths decisions, but it also determines how much suboptimal action mass remains. (Guo et al., 2019)

Finally, this sensitivity is one reason fixed-temperature softmax is usually not the preferred asymptotic choice when exact control performance matters. As with \varepsilon-greedy, a **decaying** schedule is typically needed for good long-run performance; keeping temperature fixed leaves the algorithm exposed to persistent scale mismatch and persistent soft bias. (Zhang, 2026)

Evidence

(Guo et al., 2019)

Learning Mean-Field Games

> *"Unfortunately, the discontinuity and sensitivity of argmax-e could lead to an unstable algorithm"*
>
> *"2 nstead, we consider a Boltzmann policy based on the operator softmax c : R n R n , defined as"*
>
> *"This operator is smooth and close to the argmax-e (see Lemma 7 in the Appendix). Moreover, even though Boltzmann policies are not optimal, the difference between the Boltzmann and the optimal one can always be controlled by choosing the hyper-parameter c appropriately in the softmax operator"*
>
> *".Step 2: error control in updating L. Given the sub-optimality of the Boltzmann policy, one needs to characterize the difference between the optimal policy and the non-optimal ones. In particular, one can define the action gap between the best action and the second best action in terms of the Q-value as"*
>
> *"Action gap is important for approximation algorithms (Bellemare et al., 2015), and are closely related to the problem-dependent bounds for regret analysis in reinforcement learning and multi-armed bandits, and advantage learning algorithms including A2C (Mnih et al., 2016)"*
>
> *".The problem is: in order for the learning algorithm to converge in terms of L (Theorem 2), one needs to ensure a definite differentiation between the optimal policy and the sub-optimal ones. This is problematic as the infimum of δ s (L) over an infinite number of L can be 0. To address this, the population distribution at step k, say L k , needs to be projected to a finite grid, called - net."*

(Bellemare et al., 2015)

Increasing the Action Gap: New Operators for Reinforcement Learning

> *"This paper introduces new optimality-preserving operators on Q-functions. We first describe an operator for tabular representations, the consistent Bellman operator, which incorporates a notion of local policy consistency. We show that this local consistency leads to an increase in the action gap at each state; increasing this gap, we argue, mitigates the undesirable effects of approximation and estimation errors on the induced greedy policies. This operator can also be applied to discretized continuous space and time problems, and we provide empirical results evidencing superior performance in this context. Extending the idea of a locally consistent operator, we then derive sufficient conditions for an operator to preserve optimality, leading to a family of operators which includes our consistent Bellman operator. As corollaries we provide a proof of optimality for Baird's advantage learning algorithm and derive other gap-increasing operators with interesting properties. We conclude with an empirical study on 60 Atari 2600 games illustrating the strong potential of these new operators."*

(Zhang, 2026)

A Survey of Multi-Armed Bandit Algorithms: From Theoretical Foundations to Modern Applications

> *"As with greedy, a decaying schedule must be used to achieve good asymptotic performance (Vermorel et al., 2005)."*

Known pathologies and limitations

Fixed-temperature Boltzmann exploration has several known failure modes. The main ones are: it can have unstable or non-unique fixed points, can fail to converge by standard contraction arguments, can suffer linear regret under bad or naive temperature schedules, and does not account for uncertainty in value estimates. (4 sources)

- **Multiple fixed points and instability can occur.**
  A known drawback of Boltzmann exploration is that the induced dynamics can admit **multiple fixed points**, which makes behavior sensitive to initialization and tuning. This is one reason later work proposed alternative softmax-like operators or practical hacks such as keeping the best policy seen so far to limit instability. (Ara'ujo et al., 2020)
- **The softmax Bellman operator is not generally a contraction.**
  A core pathology is that the Boltzmann / softmax operator can be **expansive** for some temperature settings, so the usual sup-norm contraction proof behind standard Q-learning convergence does not go through. This point is emphasized both in discussions of the classical Littman counterexample and in later summaries noting that softmax is not guaranteed to converge for arbitrary inverse temperatures. (Song et al., 2018)

- **Constant or poorly chosen temperature can lead to near-random behavior.**
  With low inverse temperature, softmax can stay close to uniform exploration for too long, which means it behaves almost like a random policy and can incur **roughly linear regret**. Recent bandit evidence makes this failure mode very explicit: at small inverse temperature, regret is essentially linear, while only much larger inverse temperatures make the policy resemble greedy action choice. (Overman et al., 2026)
- **Monotone temperature / learning-rate schedules can be provably bad.**
  A notable negative result from the bandit literature is that **any monotone learning-rate schedule** for Boltzmann exploration can force suboptimal behavior: the algorithm either keeps sampling bad arms too much or fails to identify the best one reliably. That work gives improved variants, but only by using more complex or problem-dependent schedules. (Cesa-Bianchi et al., 2017)
- **Good fixes may require problem knowledge that is usually unavailable.**
  Some non-monotone schedules can improve regret, but the available guarantees may require prior knowledge such as the horizon T or the gap \Delta, which limits their practical value. So even when a Boltzmann-style scheme can be repaired theoretically, the repair may depend on information standard online learners do not have. (Cesa-Bianchi et al., 2017)
- **Boltzmann exploration does not reason about uncertainty.**
  A basic limitation is that softmax over current value estimates only reacts to estimated means, not to how uncertain those estimates are. This means it can overcommit to noisy early winners or keep wasting samples on actions for the wrong reason; Cesa-Bianchi et al. identify this lack of uncertainty awareness as a crucial shortcoming of the policy. (Cesa-Bianchi et al., 2017)
- **For strong guarantees, inverse temperature often must increase over time.**
  Recent analysis again suggests that to control the “leakage” from persistent soft exploration, one needs the inverse temperature \eta_t to grow so that terms like t e^{-\eta_t \delta} are summable. This reinforces the main limitation of **fixed-temperature** softmax: keeping temperature constant leaves persistent exploratory mass on suboptimal actions and can block strong asymptotic guarantees. (Overman et al., 2026)
- **In practice, tuning itself is a pathology.**
  Even papers that use classical Boltzmann exploration describe the **temperature schedule** as the main challenge: one can hold temperature constant and tune it by search, or anneal it over time, but there is no generally robust default. This makes fixed-temperature softmax brittle across tasks and one reason practitioners often add extra safeguards or use other exploration rules. (Ara'ujo et al., 2020)

Evidence

(Ara'ujo et al., 2020)

Single-partition adaptive Q-learning

> *"Boltzmann exploration has its drawbacks, namely the existence of multiple fixed points (Littman, 1996). To deal with this issue, alternative softmax-based operators have been proposed by (Asadi et al., 2016) and Pan et al. (2019). The approach followed in this paper uses the classical softmax function, while keeping track of the best policy found so far, therefore clipping the instability associated with Boltzmann exploration"*

> *".The main challenge involved with the use of Boltzmann exploration is setting the temperature schedule. A straightforward approach, used by (Tijsma et al., 2016), is to set a constant temperature, which can be found using grid search or any other tuning method. Another approach is to decrease (anneal) the temperature as some function of the training iteration (Pan et al., 2019)."*

(Song et al., 2018)

Revisiting the Softmax Bellman Operator: Theoretical Properties and Practical Benefits

> *"one commonly-used strategy is to add randomness by replacing the max function with the softmax function, as in Boltzmann exploration (Sutton and Barto, 1998)"*

".it has also been demonstrated that the softmax Bellman operator is not a contraction, for certain temperature parameters (Littman, 1996)"

".Finally, although the softmax operator is not guaranteed to converge for any inverse temperature parameters"

"Littman (1996) first showed that the Boltzmann operator could be expansive in a specific MDP by tuning the temperature parameter."

(Overman et al., 2026)

Annealed Softmax Greedy in Many-Armed Bayesian Bandits

"Theorem 5.2 requires η t → ∞ (i.e. increasing inverse temperature over time) to make the leakage term t exp(−η t δ) summable"

".with few arms, all empiricalmean-based methods accumulate roughly linear regret"

"At low inverse temperature (η = 1), the Softmax policy explores nearly uniformly and accumulates regret linearly, behaving essentially as a random policy. As η increases, the policy concentrates mass on arms with higher empirical means, and regret curves flatten toward the Greedy baseline. By η = 50, the Softmax algorithm is nearly indistinguishable from empirical-mean greedy"

".By contrast, with a small number of arms, the same kind of softmax policy can suffer linear regret."

(Cesa-Bianchi et al., 2017)

Boltzmann Exploration Done Right

"Despite its widespread use, there is virtually no theoretical understanding about the limitations or the actual benefits of this exploration scheme. Does it drive exploration in a meaningful way? Is it prone to misidentifying the optimal actions or spending too much time exploring the suboptimal ones? What is the right tuning for the learning rate?"

"One of our main results is showing that the Boltzmann exploration strategy with any monotone learning-rate sequence will induce suboptimal behavior. As a remedy, we offer a simple non-monotone schedule that guarantees near-optimal performance, albeit only when given prior access to key problem parameters that are typically not available in practical situations (like the time horizon T and the suboptimality gap Δ). More importantly, we propose a novel variant that uses different learning rates for different arms, and achieves a distribution-dependent regret bound of order $\frac{K\log^2 T}{\Delta}$ and a distribution-independent bound of order $\sqrt{KT}\log K$ without requiring such prior knowledge"

".We first show that any monotone learning-rate schedule will inevitably force the policy to either spend too much time drawing suboptimal arms or completely fail to identify the optimal arm. Then, we show that a specific non-monotone schedule of the learning rates can lead to regret bound of order K log T Δ 2 . However, the learning schedule has to rely on full knowledge of the gap Δ and the number of rounds T . Moreover, our negative result helps us to identify a crucial shortcoming of the Boltzmann exploration policy: it does not reason about the uncertainty of the empirical reward estimates."